# COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 5

## 1 Computer Networking (awm22)

Consider a satellite network consisting of a constellation of $S$ satellites placed at an altitude of 1,000km above the Earth (Earth's diameter is ≈12,750km) and $G$ ground stations connecting both customers and Internet exchange points. Each orbit of the constellation consists of 50 satellites (20 orbits total). Each satellite is equipped with five communications laser-receivers; one to communicate with the ground and four to communicate with the nearest neighbour satellites. The satellites are not geostationary. You may assume each satellite only communicates with its nearest neighbours at any time. Each node—satellite, or ground station—has a unique identifier of 16 bytes which it knows.

Design a topology-discovery protocol than can identify the shortest path among any two ground stations using the 1,000 satellites in orbit.

Symmetric paths may be presumed.

switch/network topology discovery

(*a*) Outline a protocol (including message formats) for a node to learn about its immediate neighbours. [3 marks]

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*Answer:*

There are numerous solutions in this space, here is one.

Each node sends out a message of the form

```
  |  TOPOLOGY PROBE |  NODE ID  |
```

each node receiving such a message replies with

```
  |  TOPOLOGY REPLY |  NODE ID  |
```

using its own NODE ID

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(*b*) Design a protocol (including message formats) for distributing this information across the network. [7 marks]

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*Answer:*

There are numerous solutions in this space, here is one:

In order to build up the topology, it is not enough to simply exchange information with immediate neighbours (aka next-hop table).

each node in the network must receive neighbour information for every other node.

For example a k (maximum degree) entry message of the form.

```
NEIGHBOURS  NODE ID
Neighbour 1
Neighbour 2
...
Neighbour k (max)
```

would be flooded across the entire constellation with a time to live count of d (diameter); an estimate of the diameter is the half-circumference of any single ring (25 satellites).

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($c$) Give a bound on the total amount of non-redundant information which is transmitted to ensure that every node acquires complete topology information. [5 marks]

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*Answer:*

for the example above two phases: one is the immediate discovery of neighbours; one each of the two messages of the protocol of part ($a$) $O(5 \times theMessageSize \times (1000 satellites + nGroundStations)$

the second is the (flooding) of per-node connectivity where the minimum information exchange is given as $5 \times theMessageSize \times (1000 satellites + nGroundStations)$

as this is an order computation an order-of-magnitude estimate will be acceptable but the candidate will only get full marks if they recognise the ground stations must be included as nodes within the computation. The specification of non-redundant is intended to stop students working out the number of circuits any level of convergence might take.

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($d$) The channel bandwidth is given as $B$, an approximation of $3 \times 10^8$ meters per second may be used for the speed of light, and per node packet processing time may be considered zero.

($i$) Make an estimate of the worst-case total amount of time the exchange of this information will take to propagate across the network. [2 marks]

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*Answer:*

useful numbers
50 nodes per orbit
approx 400km between each satellite so thats about 1.3ms one way
worst case dnd to each is 2 ground links

This will depend critically upon the solution they propose to Part ($b$) however the answer is usually considerably more trivial than first appears: an upper bound is $O(|V| \cdot |E|)$

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($ii$) Outline one method to improve the time-period. Speculate on the improved upper bound of total time for the information exchange. [3 marks]

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*Answer:*

Several answers are possible, for example

We know the fixed structuring of the rings in particular we possess stabile knowledge of our own ring, eg our neighbours within that ring; this immediately reduces the adjacencies from 5 to 3. Other structural regularities will also mean the problem is reduced considerably in scope/complexity. Solutions to finding paths becomes an exercise in utilising such structural knowledge to constrain the number of paths requiring computation.

If a shortest path method is used and adjacencies within the same orbit in space can be precomputed (reducing the flood / reducing the need to learn these paths). Orbital mechanics would allow all orbits to be known and predictable reducing the need to continuously learn these paths.

For the per-orbit case (reducing the adjacencies from 5 to 3); the total number of edges is reduced to three fifths of $O((|\frac{3}{5} \cdot V_{satellite} + V_{groundstation})| \cdot |E|)$ (approximately)

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